

Robin Lange

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EDUCATION

Bachelor of Science in Mechanical Engineering

Georgia Southern University • Statesboro, GA • May 2027 • GPA: 4.0

President's List, Allen E. Paulson Scholar, Honors College Scholar

SKILLS

ML and AI: PyTorch, ONNX, Deep Learning (LSTM, YOLO), Computer Vision, Object Detection, OpenSCENARIO 1.3, ADAS Scenario Design, Behavior Trees.

Robotics: ROS2, Visual SLAM, Nav2, Gazebo, MAVROS, ArduPilot, Sensor Fusion, Motion Planning, Simulation and Testing

Software and Hardware: Python, C++, MATLAB, Linux, Docker, Git – NVIDIA Jetson Orin, Ouster LiDAR, Radar, Intel RealSense Camera

Languages: German (Native), English (Fluent), French (Intermediate)

EXPERIENCE

Student Researcher

Intelligent Vehicles Navigation Research Laboratory Georgia Southern University

October 2023—Present, Statesboro, GA

- Designing AEB validation scenarios in OpenSCENARIO 1.3 within AVL Scenario Simulator, covering CCRb, CCRm, and CCRs protocols across car and truck variants with parameterized speed and following-distance configurations.
- Developed a warehouse robot platooning framework (leader-follower AMRs, PRM path planning, envelope-based collision prevention). 20-trial simulation cut static-obstacle collisions by 88%.
- Built a cuVSLAM Visual SLAM pipeline on NVIDIA Jetson Orin Nano with Intel RealSense D455 using Isaac ROS / ROS2 Humble, validated against dual-beacon ground truth. Achieved MAE 8.8 cm, RMSE 14.5 cm, matching NVIDIA KITTI benchmark.
- Wrote a data-driven sim-to-real calibration paper: PyTorch LSTM on 87K+ encoder samples, ONNX export, deployed in Gazebo ROS2 plugin. Reduced simulation RMSE 36%, MAE 29% vs. baseline.

Intern for Vehicle Tests

CARISMA Institute of Safety in Future Mobility (C-ISAFE)

July 2025—August 2025, Ingolstadt, Germany

- Trained and evaluated a custom YOLOv6l model in PyTorch for truck-rear detection on BDD100K, benchmarked against a TFLite YOLOv5 baseline. Quantified per-class detection gaps to support ADAS sensor integration decisions.
- Designed and validated a soft surrogate crash target for the KICSaFe project, replicating truck-rear geometry to enable repeatable multi-modal ADAS sensor evaluation across underwrite collision scenarios.
- Analyzed multi-modal sensor return signals (radar, LiDAR, camera) across approach angles and distances to extract truck-rear scattering signatures, providing quantitative parameters for the surrogate target's electromagnetic and optical profile.

Autonomous Systems and AI Intern

Fraunhofer Institute for Transportation and Infrastructure Systems IVI

June 2024—August 2024, Ingolstadt, Germany

- Engineered a novel autonomous UAV control system using Behavior Trees within ROS2, benchmarked against a legacy State Machine architecture on mission-critical flight sequences. Selected BT approach for production based on a 20% latency reduction and superior modularity.
- Empirically evaluated 3 control algorithm variants for resource efficiency on embedded UAV hardware. Documented performance bounds and failure modes for production handoff; received an outstanding evaluation.

SELECTED PUBLICATIONS AND PRESENTATIONS

Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing

Co-Author • SAE Technical Paper 2026-01-0058 • WCX World Congress • April 2026

Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underwrite Testing

Lead Author • journal paper in preparation

Warehouse Platooning Framework using a Sensor-Rich Leader and Minimal Perception Follower

Co-Author • Submitted, Control Engineering Practice • 2026

Intelligent Indoor UAV Navigation with V-SLAM and Behavior Trees for Industry 5.0 Applications

Presenter • Southern Regional Honors Council Conference • March 2026

Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications

Co-Author • SAE Technical Paper 2025-01-8025 • WCX World Congress • April 2025

Drone Guiding through Equivalent Hand Movement Transversion

Presenter • 2nd Place, Georgia Power Innovation Award • CEC Research Symposium • April 2025

INVOLVEMENT

Southern Leaders, Tau Beta Pi, SAE International, Honors Student Organization, Engineering and Physics TA, FYE Peer Leader